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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																												
B									B																																												
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-23 20kV Compressor Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																												
D									D																																												
E									E																																												
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="3"> ABB EPMV PT. ABB Sakti Industri </td><td rowspan="3">COVER SHEET SLD & GA Drawing MVD-23 20kV Compressor</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB EPMV PT. ABB Sakti Industri	COVER SHEET SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.		Cont. 16	F
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SHEET INDEX

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
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									Drawing Nbr.		Sh.	2
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A																																																				
B	SWITCHGEAR VERSION				= UNISEC				B																																																				
	RATED VOLTAGE (Un)				= 24 kV																																																								
	OPERATING VOLTAGE				= 20 kV																																																								
	RATED FREQUENCY (fr)				= 50 Hz																																																								
	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)				= 125 kV																																																								
C	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)				= 50 kV				C																																																				
	RATED CURRENT OF MAIN BUSBARS (Ir)				= 630 A																																																								
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)				= 20 kA																																																								
	RATED PEAK WITHSTAND CURRENT (Ip)				= 50 kA																																																								
	RATED DURATION OF SHORT CIRCUIT (tk)				= 1 s																																																								
D	INTERNAL ARC CLASSIFICATION (IAC)				= AFLR / 16kAx1s				D																																																				
	AMBIENT CONDITION				= NORMAL																																																								
	AMBIENT AIR TEMPERATURE				= -5°C...+40°C																																																								
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)				= IP3X																																																								
	DEGREE OF PROTECTION WITH OPEN DOORS				= IP2X																																																								
E	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)				= 230VAC				E																																																				
	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)				= 230VAC																																																								
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)				= 230VAC																																																								
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS				= 450/750V																																																								
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS				= PVC																																																								
F	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS				= 2.5 mm²/BLACK				F																																																				
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS				= 1.5 mm²/BLACK																																																								
	CROSS-SECTION OF CONDUCTORS FOR MOTORS				= 2.5 mm²/BLACK																																																								
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)				= 1.5 mm²/BLACK																																																								
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS				= 1.5 mm²/BLACK																																																								
CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE				= 2.5 mm²/BLACK																																																									
COMMUNICATION CABLE				= N/A																																																									
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS														A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)														
B	DESIGNATION	DESCRIPTION													B
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)													
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1													
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2													
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3													
	-BAR	VOLTAGE PROTECTION RELAY													
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1													
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2													
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3													
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1													
C	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2													C
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3													
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1													
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2													
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3													
	-BCD	DIFFERENTIAL PROTECTIVE RELAY													
	-BCF	FEEDER PROTECTION RELAY													
	-BCG	GENERATOR PROTECTION RELAY													
	-BCM	MOTOR PROTECTION RELAY													
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER													
D	-BCP	TRANSFORMER PROTECTION RELAY													D
	-BCR	CURRENT PROTECTION RELAY													
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1													
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2													
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3													
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2													
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3													
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2													
E	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3													E
	-BCZ	DISTANCE PROTECTION RELAY													
	-BEF	FREQUENCY PROTECTION RELAY													
	-BER	SUPERVISION RELAYS													
	-BES	SYNCHRONIZING RELAY													
	-BET	THERMAL PROTECTION RELAY													
	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT													
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT													
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
F	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER													F
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS													

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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION		-CA		CAPACITORS										
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION		-CA1		CAPACITOR FOR FRONT VENTILATOR										
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		-CA2		CAPACITOR FOR REAR VENTILATOR										
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION		-EA1		LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.										
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION		-EA2		LIGHTING LAMP LOCATED IN CABLE COMPARTMENT										
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION		-EA3		LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE										
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION		-EA4		LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT										
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION		-EB1		HEATER LOCATED IN CABLE COMPARTMENT										
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION		-EB2		HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT										
	-BGK		POSITION SWITCH OF KEY LOCK		-EB3		HEATER LOCATED IN MOTOR CABINET										
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1		-EB4		HEATER FOR VENTILATION TEST										
	-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS		-EB5		HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.										
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION		-EB7		HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE										
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION		-EB8		HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE										
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION		-FA1		SURGE ARRESTER LOCATED ON PHASE L1										
	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION		-FA2		SURGE ARRESTER LOCATED ON PHASE L2										
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION		-FA3		SURGE ARRESTER LOCATED ON PHASE L3										
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION		-FCD		FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS										
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION		-FCF1		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1										
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION		-FCF2		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2										
C	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION		-FCF3		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3										
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION		-FCM1		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER										
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION		-FCM2...-FCM10		MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS										
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR		-FCM11		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1										
	-BGV2		POSITION SWITCH OF REAR VENTILATOR		-FCM12		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2										
	-BM		HYGROSTAT		-FCM13		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3										
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-FCT		THERMAL OVERLOAD RELEASES										
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GA		GENERATORS										
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ1		FRONT VENTILATOR										
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ2		REAR VENTILATOR										
D	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER		-GQ3		VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK										
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2		-KFA1		AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE										
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3		-KFA2		AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE										
	-BR		FLAME DETECTORS, SMOKE DETECTORS		-KFA3....-KFA9		AUXILIARY RELAYS OR CONTACTORS										
	-BT		THERMOSTAT		-KFC		CLOSING RELAYS OR CONTACTORS										
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS		-KFI		INTEGRATED CIRCUITS										
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS		-KFL		LOCKOUT RELAY										
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM		-KFM		MICROPROCESSORS										
	-BU		BUCHHOLZ RELAY		-KFN		ANTIPUMPING RELAYS										
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING		-KFO		OPENING RELAYS OR CONTACTORS										
E	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING		-KFP		PROGRAMMABLE LOGIC CONTROLLERS (PLC)										
	-BZ,-BZ4		MULTIFUNCTION SENSORS		-KFR		RECLOSING RELAYS										
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1		-KFS		SYNCHRONIZING DEVICES										
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS										
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3		-KFZ		CONTROLLERS										
					-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER										
					-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD										
					-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1										
					-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2										
					-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS										
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

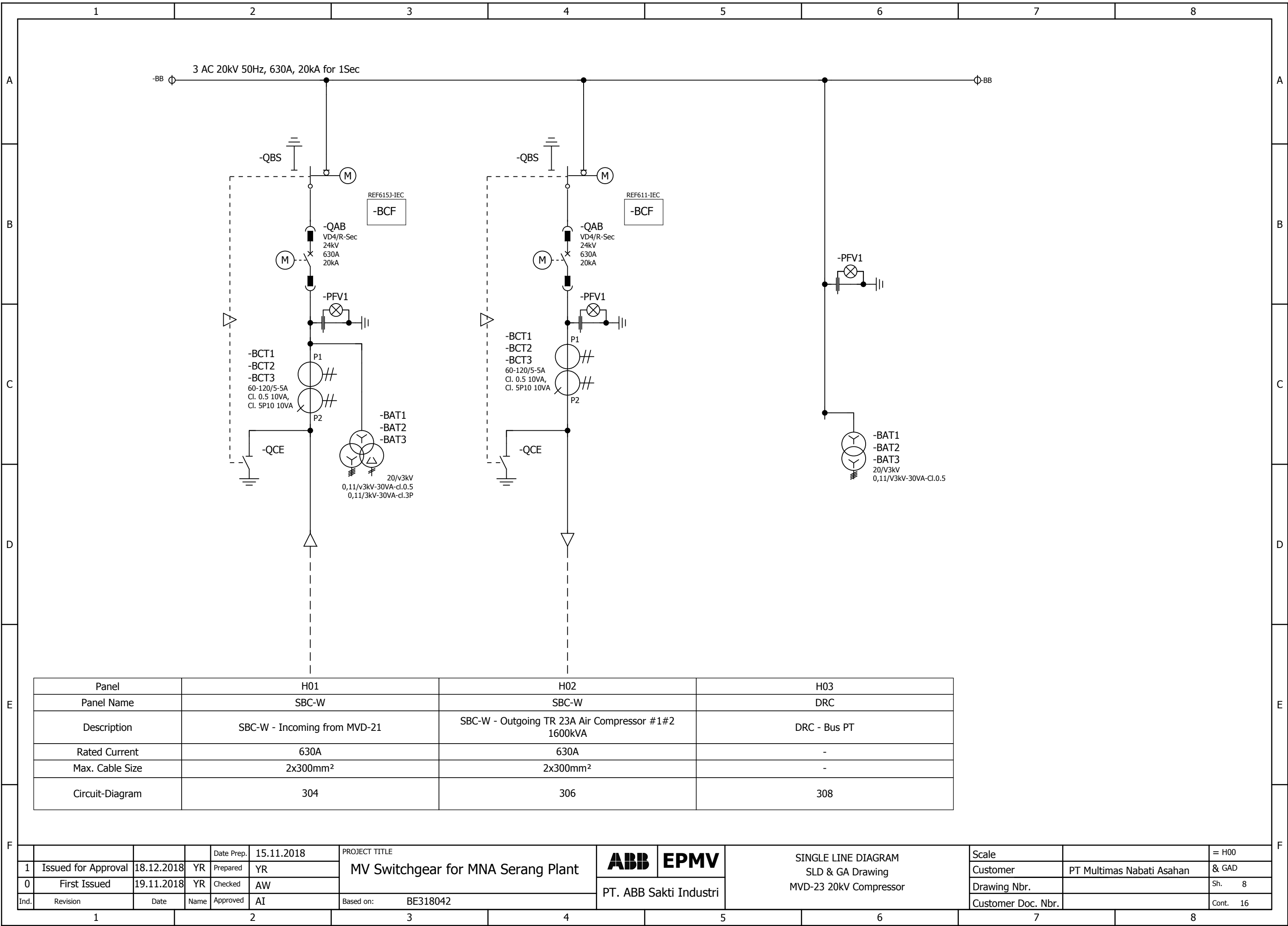
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			Drawing Nbr.			Sh. 6		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont. 16		

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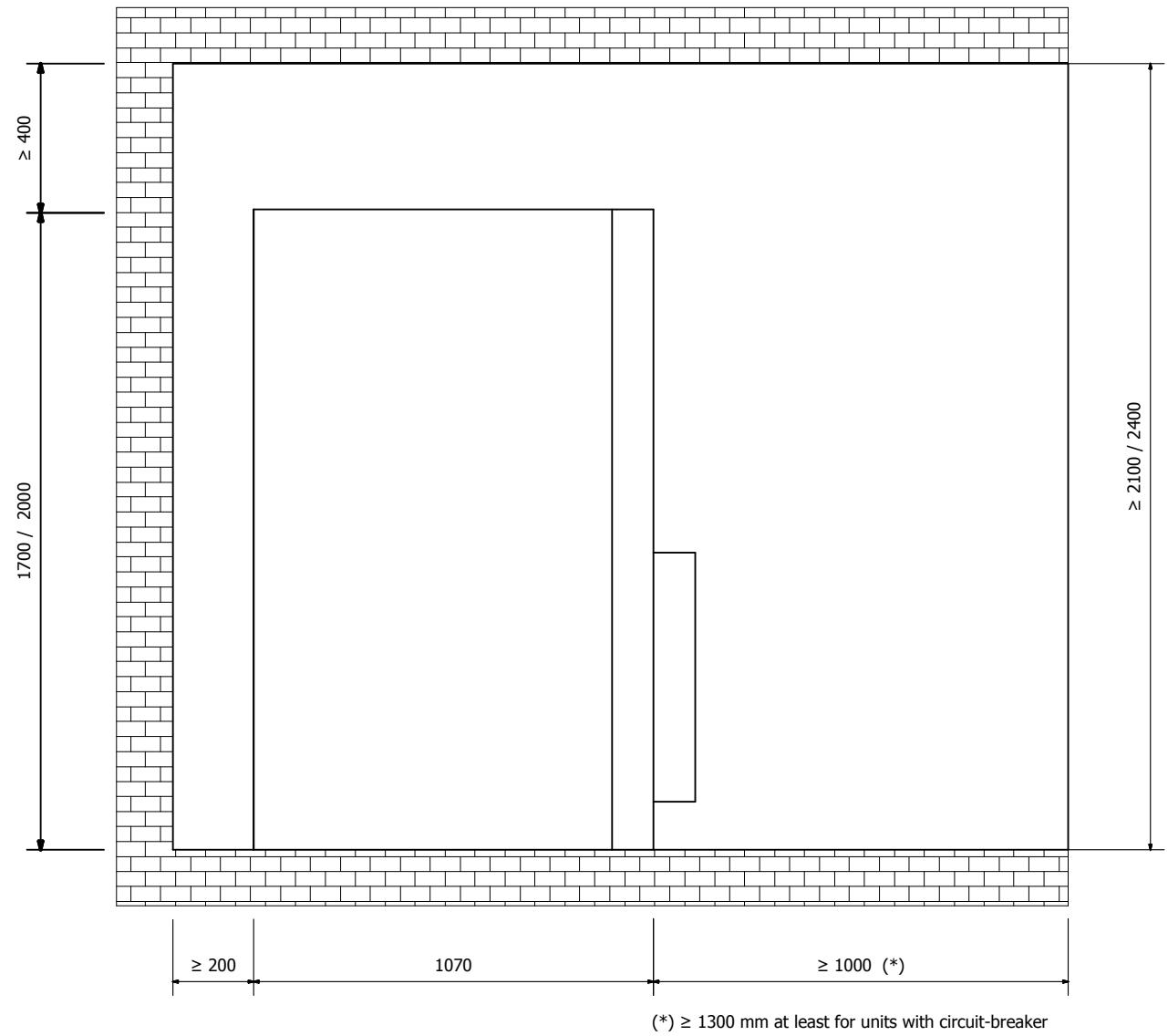
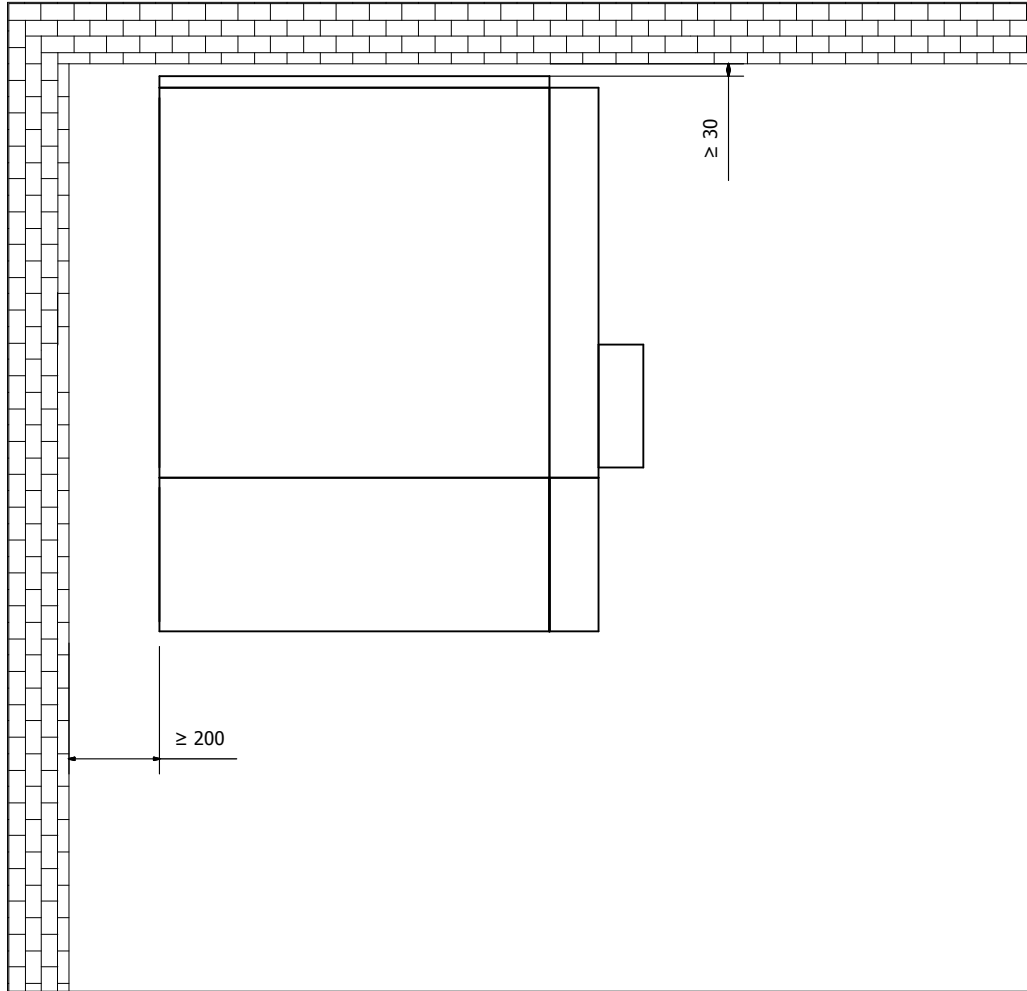
A	-TFA		ACTIVE POWER TRANSDUCERS		-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		A																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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	-TFF		FREQUENCY TRANSDUCERS		-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-TFJ		ACTIVE ENERGY TRANSDUCERS			-XD14			TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-TFK		REACTIVE ENERGY TRANSDUCERS		-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-TFM		MULTIFUNCTION TRANSDUCERS			-XD17			TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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	-TFQ		REACTIVE POWER TRANSDUCERS			-XD19			TERMINAL BLOCK FOR SPECIAL USE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
B	-TFS		SIGNAL CONVERTERS		-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD		B																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
	-TFV		VOLTAGE TRANSDUCERS			-XDJ1			TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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	-WBC		POWER CABLES			-XDS			CURRENT SOCKETS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			-XDV2			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			-XDV4			CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XDX	SUPPORT TERMINAL BLOCK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			-XE1			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			-XE5			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-XDB10...-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
	-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS			-XE7			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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	-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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	-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
	-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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	-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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RULES FOR INSTALLATION IN THE ROOM

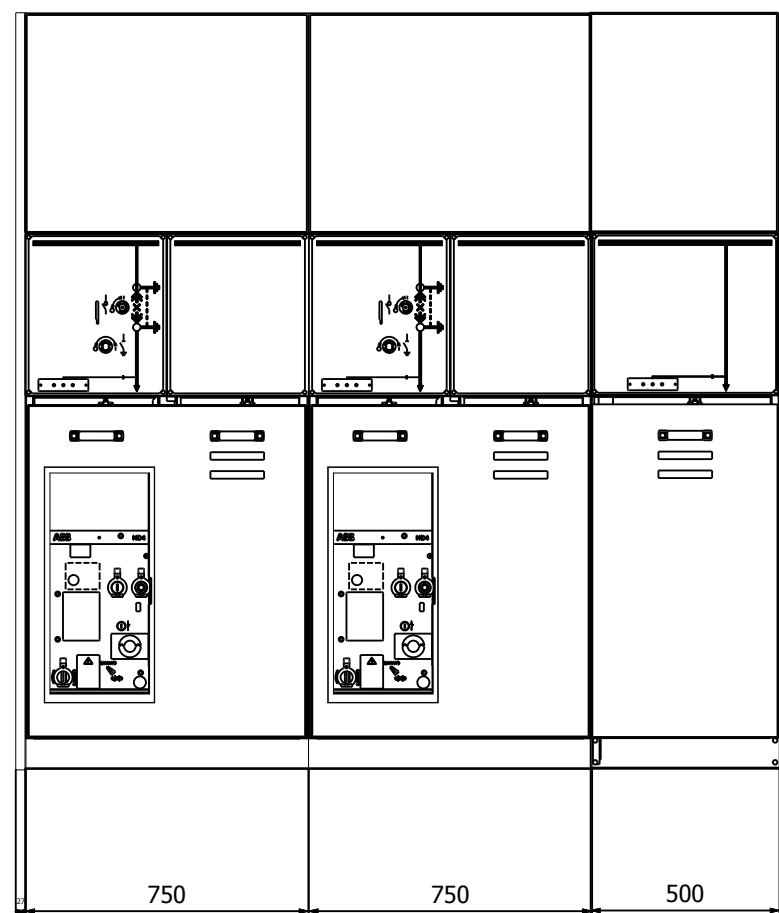
MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	 	INSTALLATION RULES SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR				Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW				Drawing Nbr.		Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 16

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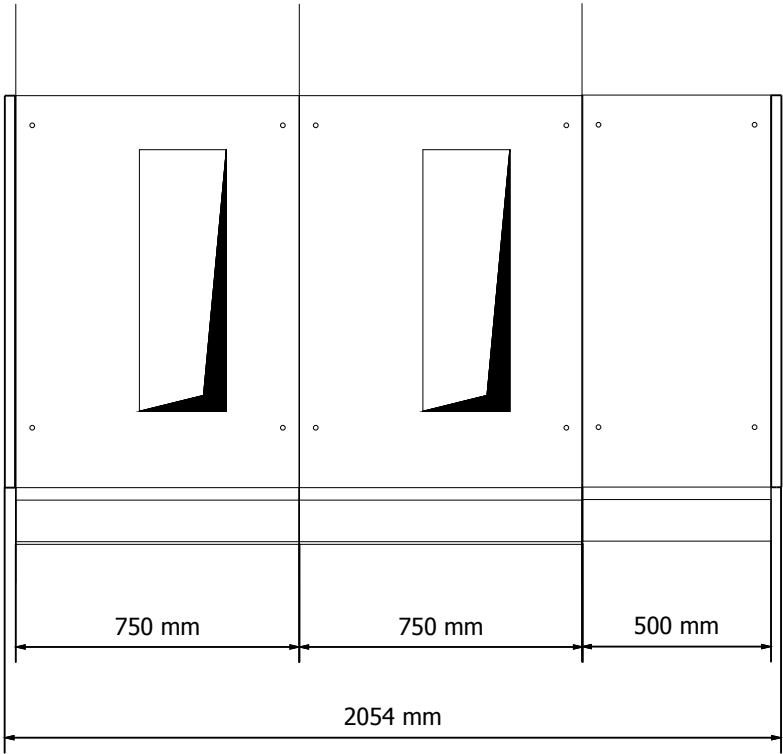


Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 10	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.				Cont. 16

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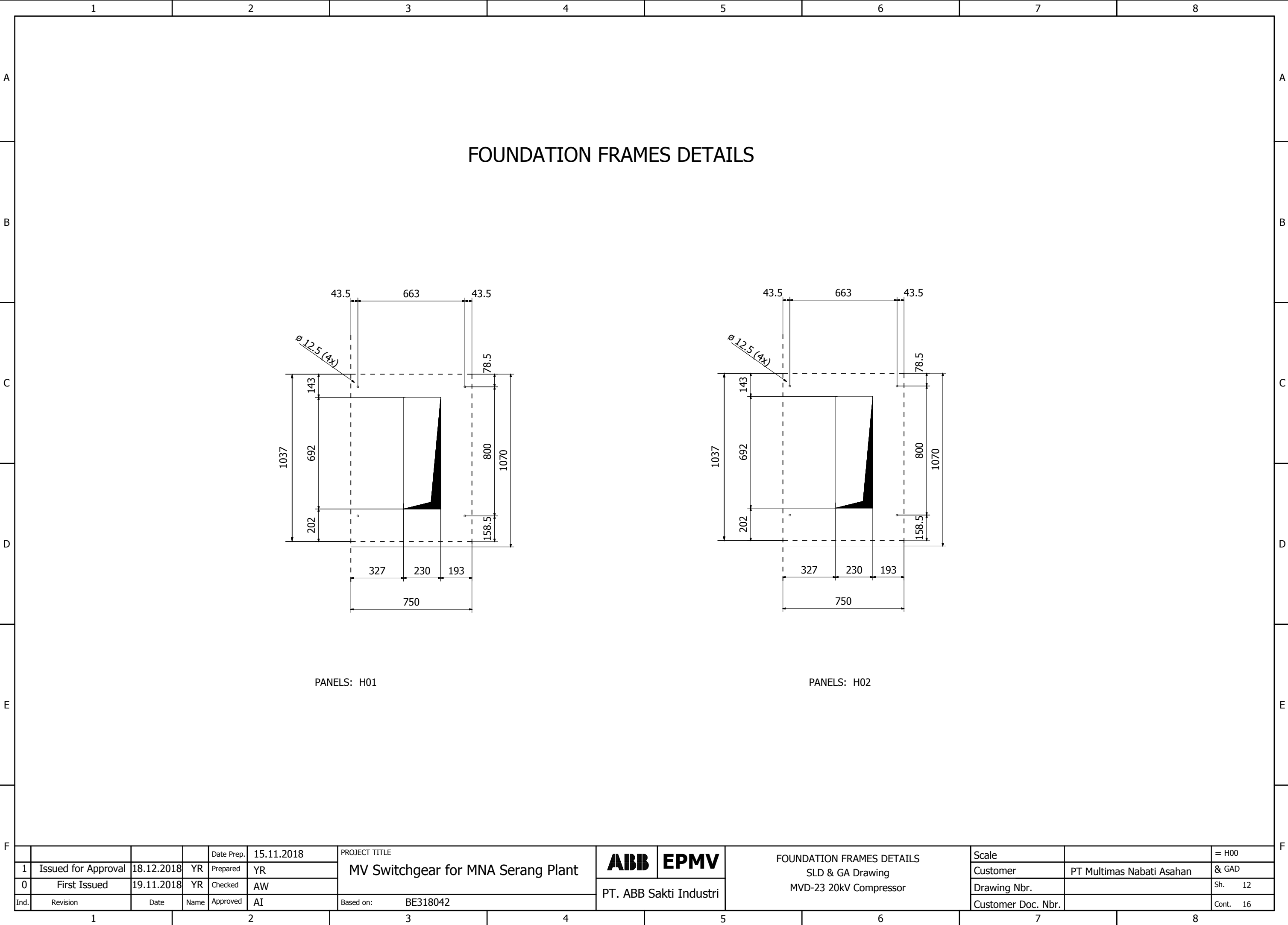
FOUNDATION FRAMES



Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 11		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 12		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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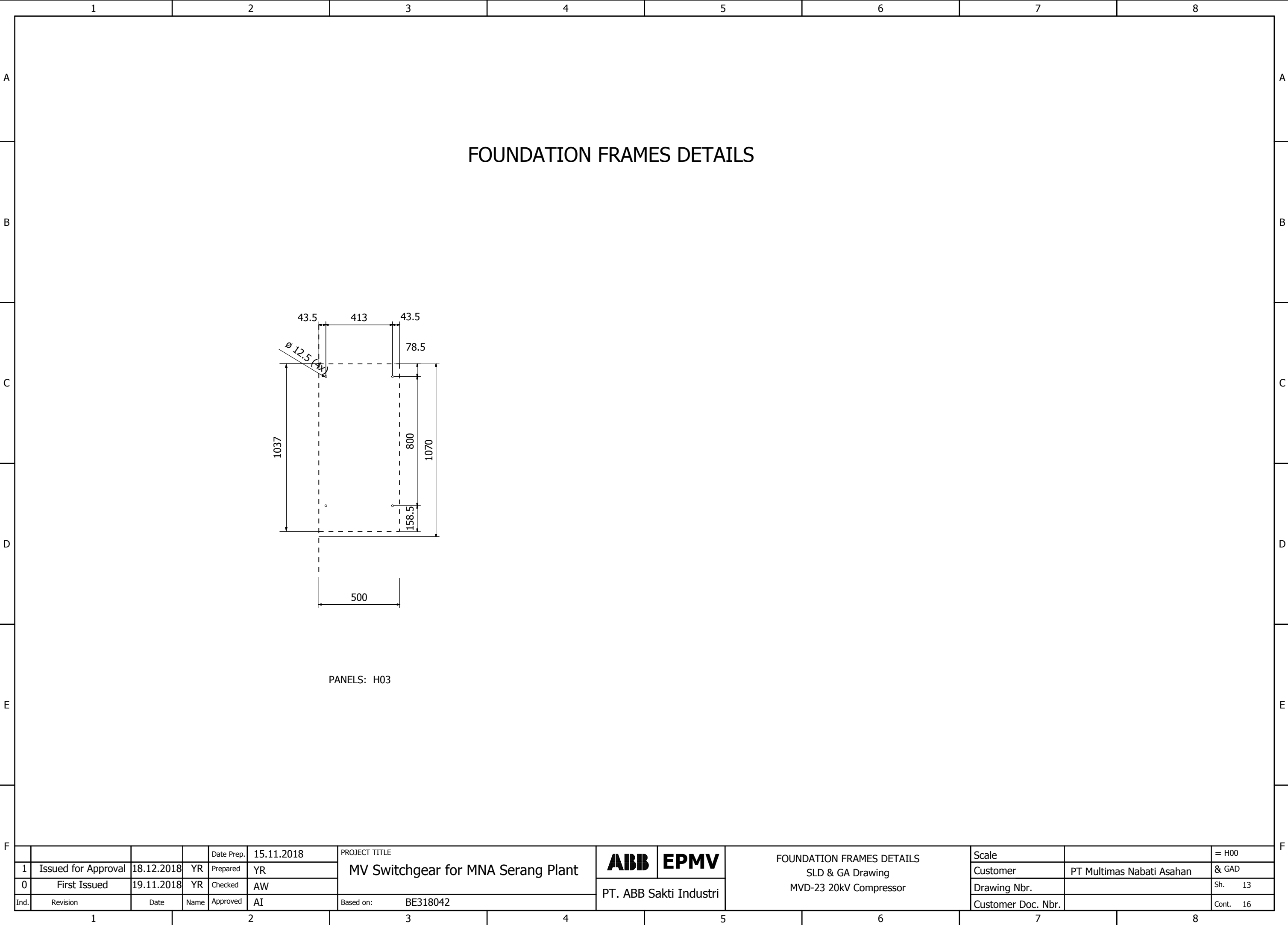
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 13		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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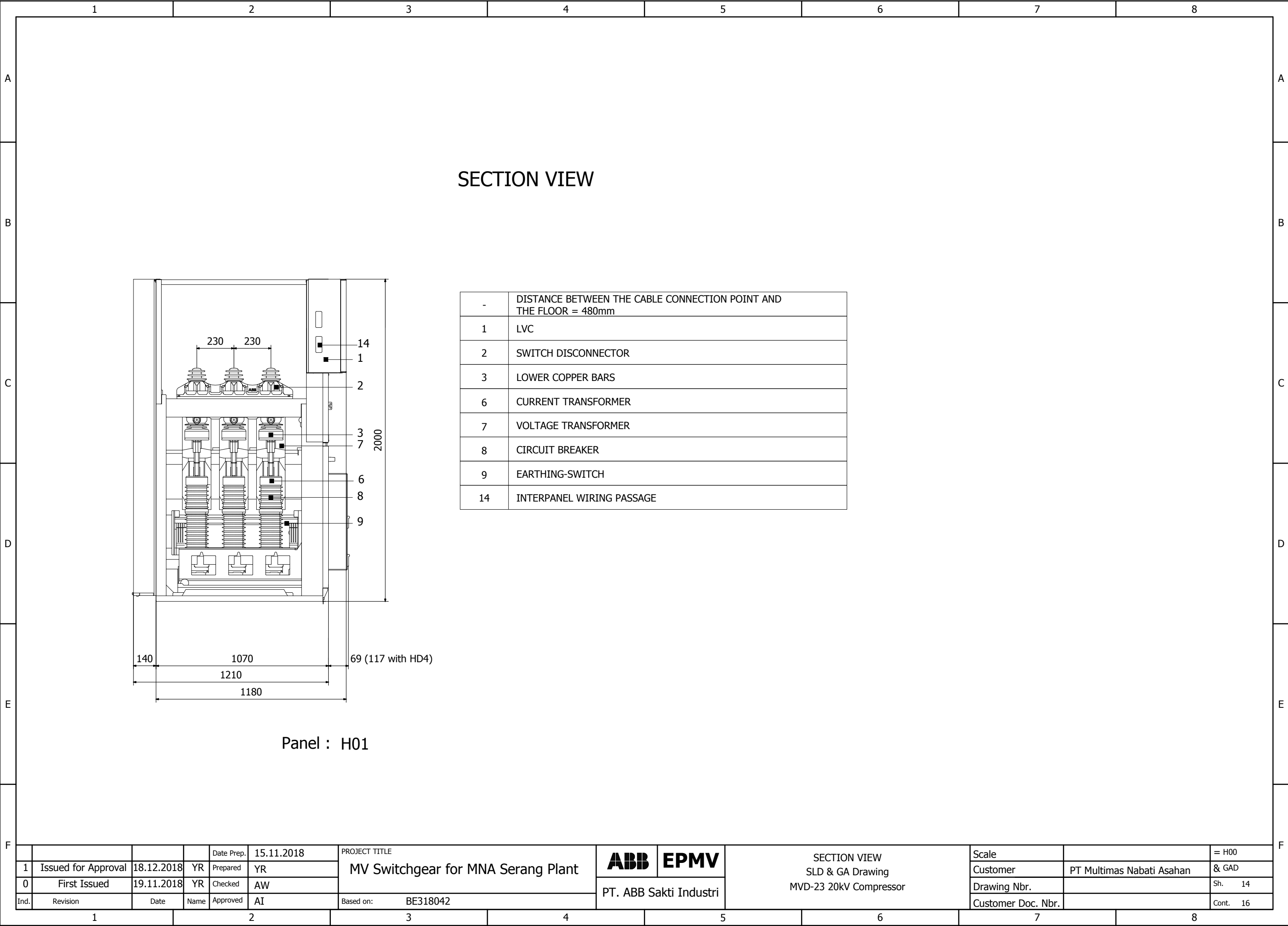
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			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared						YR	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked			AW	Drawing Nbr.			Sh. 14	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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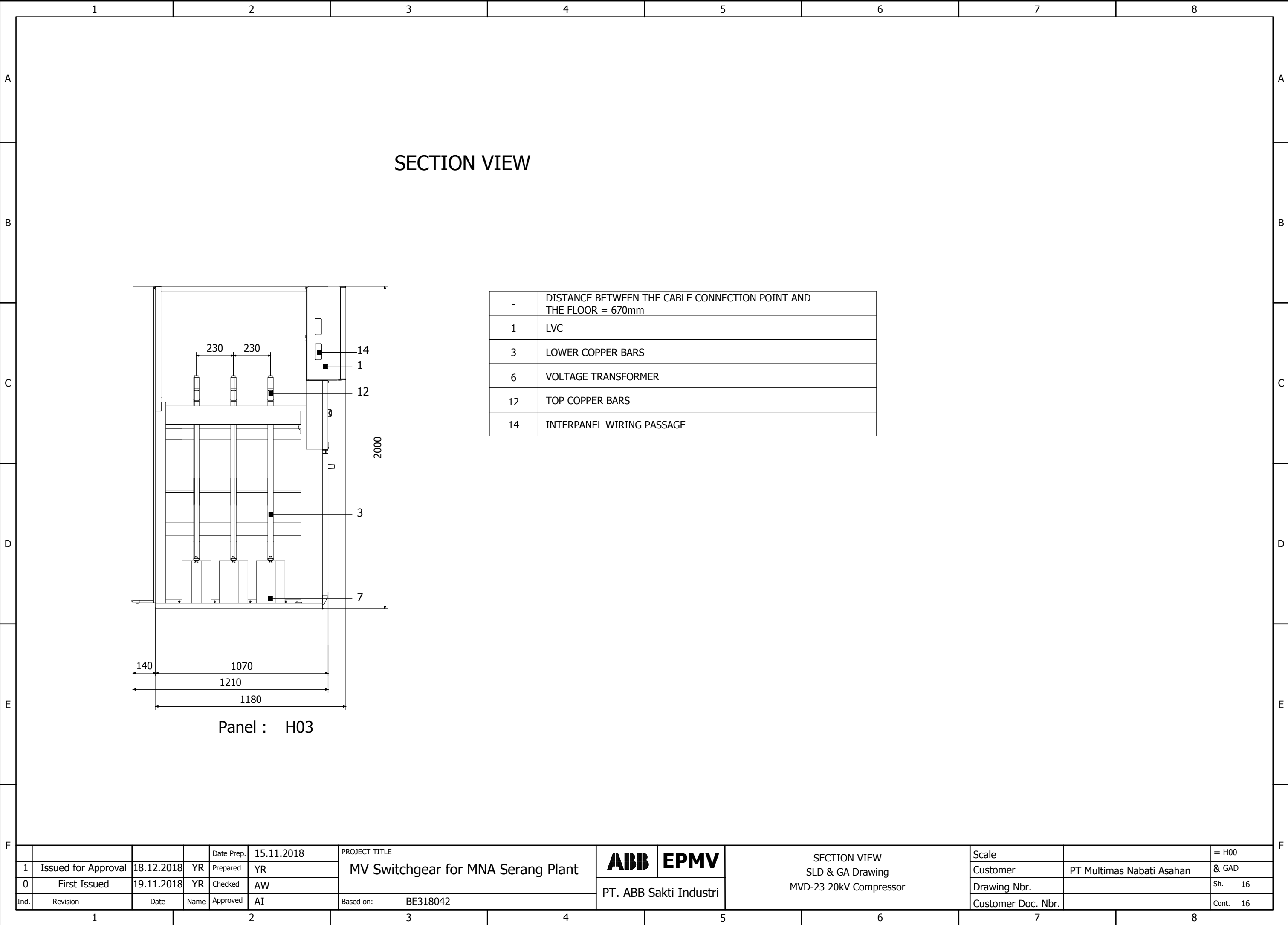
F

-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 480mm
1	LVC
2	SWITCH DISCONNECTOR
3	LOWER COPPER BARS
6	CURRENT TRANSFORMER
8	CIRCUIT BREAKER
9	EARTHING-SWITCH
14	INTERPANEL WIRING PASSAGE



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri		
0	First Issued	19.11.2018	YR	Checked	AW							
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042					
								Customer			PT Multimas Nabati Asahan	& GAD
									Drawing Nbr.		Sh.	15
									Customer Doc. Nbr.		Cont.	16

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-23 20kV Compressor	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 16		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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